Quantum computing @ LINKS Foundation

QCS-Lab, HPC integration, applications

PAOLO VIVIANI - LINKS CPE

SEPTEMBER 10TH 2026 - SCHOOL ON QUANTUM SIMULATION – RPMBT23

LINKS FOUNDATION

A 20-year history marked
by continuous growth and evolution

INNOVATION

KNOWLEDGE

LINKS FOUNDATION IS AN INSTRUMENTAL BODY OF
COMPAGNIA DI SAN PAOLO

AND OPERATES AS AN INSTRUMENTAL BODY OF
POLITECNICO DI TORINO

80%
EFFORT:
APPLIED RESEARCH

20%
EFFORT:
FRONTIER RESEARCH

€19M+
TURNAROUND 2026

185+
RESEARCHERS

35%

65%

Average age 37,7

12 spoken languages

SPECIALIZATIONS
engineering, physics, mathematics, computer science, psychology, economics, communication science, architecture.

COUNTRY OF ORIGIN
Our researchers come from all over the world.

TEAM Quantum Computing

Advanced Computing

Giacomo Vitali
Senior Researcher/Team Leader

Dr. Olivier Terzo
CPE Domain Head

Politecnico di Torino

Prof. Montrucchio
Full Professor @ PoliTo - DAUIN

Alberto Scionti
Senior Researcher

Chiara Vercellino
Senior Researcher

Paolo Viviani
Senior Researcher

Emanuele Dri
Senior Researcher

+

PhDs and Internships/Master Thesis students from PoliTo and UniTo

MIT Agreement

Massachusetts Institute of Technology

IQM Spark machine

International positioning

Collaborations

Pawsey
QuEra Computing Inc.


"Lagrange" Quantum Tech Piedmont Initiative
FONDAZIONE
links
PASSION FOR INNOVATION
1859
Politecnico
di Torino
INRiM
ISTITUTO NAZIONALE
DI RICERCA METROLOGICA

Lagrange: identikit

- ✓ Machine name: **Lagrange**
- ✓ Model: **IQM Spark**
- ✓ Vendor: IQM
- ✓ Size: **5 qubits**
- ✓ Technology: **superconducting**

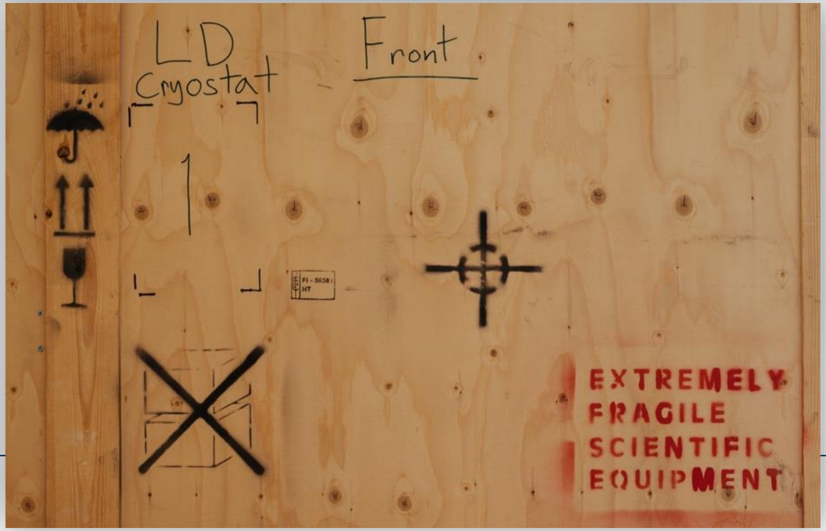
Acquired by: **QTech Piemonte Strategic Initiative**

FONDAZIONE LINKS PASSION FOR INNOVATION

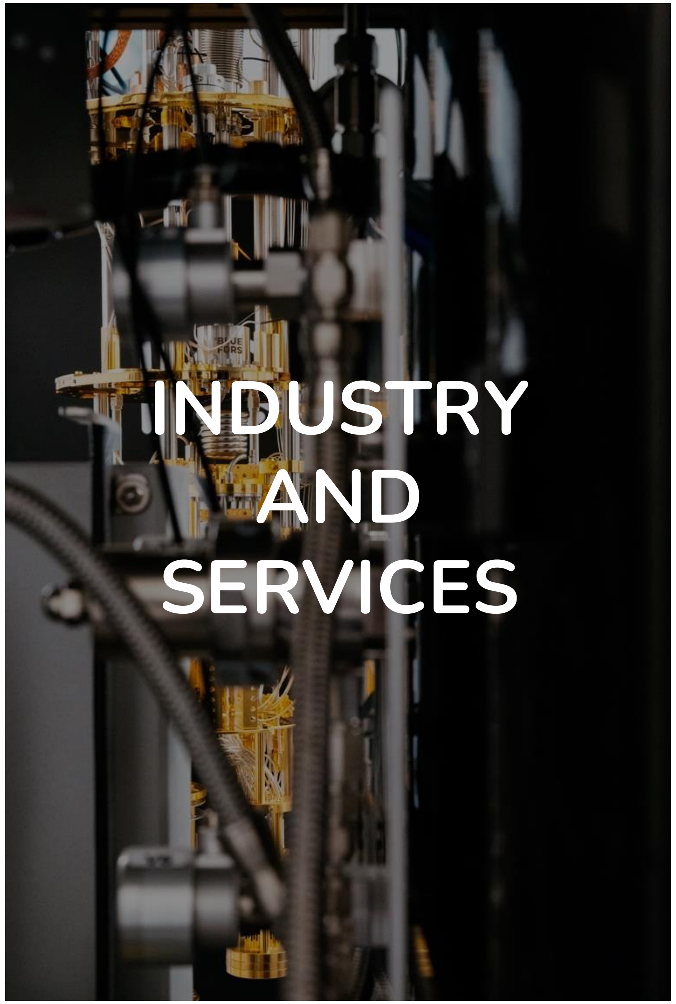
Politecnico di Torino 1859

INRiM ISTITUTO NAZIONALE DI RICERCA METROLOGICA

22 May 2025: first IQM quantum computer in Italy



Three pillars



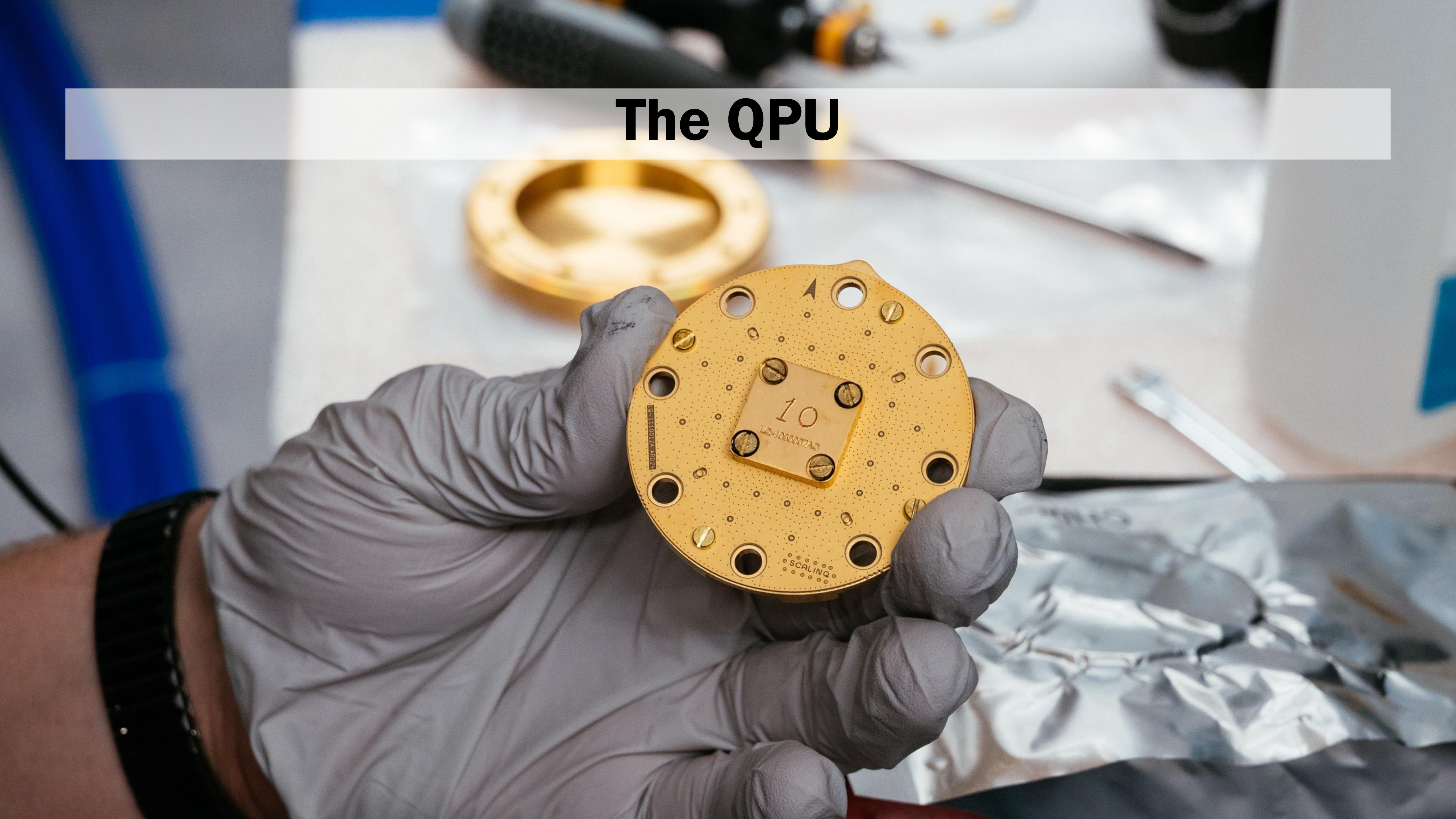
QCS-Lab

QUANTUM COMPUTING AND SIMULATION LAB

- As part of the **Regional Strategic Initiative on Quantum Technologies**, LINKS acquired an **IQM Spark machine** named Lagrange
- Machine operational since September 2025
- Used by LINKS Reseachers, Politecnico students (**including classes**) and faculty, INRIM researchers
- **> 750k jobs** so far
- **~ 2 QPU week (!)** since September
- **> 800 users**
- **Very first HPC-QC hybrid job fully executed in Italy** (Leonardo+ Lagrange)

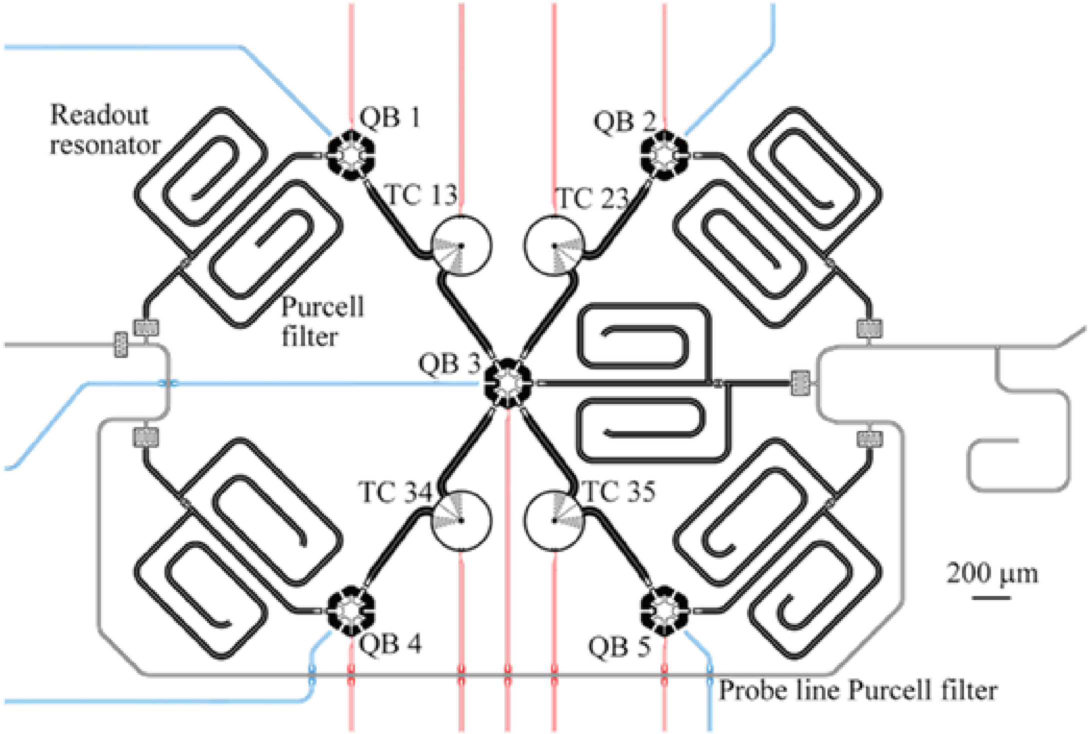
HARDWARE

The QPU



Quantum Processing Unit (QPU) Overview

[On-premises superconducting quantum computer for education and research, J. Rönkkö et al, EPJ Quantum Technology, April 2024]



- Implements quantum computation using **superconducting** qubits and couplers
- IQM **Spark** prototype QPU: **5** data qubits in a star topology
- Each **qubit** is a **transmon**, a tiny superconducting circuit made of a **Josephson junction** (which provides quantum nonlinearity) and a **large shunt capacitor**.
- Each qubit can be **individually controlled by microwave signals** to perform **rotations** and **logic gates**.

Specifications

Reference values reported by IQM for the **IQM Spark**

- **systematic characterization campaigns**,
- performed across **multiple devices**,
- under **standard operating conditions**.

Floor footprint	125 x 315 cm
Minimum ceiling height	290 cm
Minimum load bearing capicity	800 kg/m^2
Typical mean total electrical power consumption	11 kW

Metric	Minimum	Typical
Median single-qubit gate fidelity	≥ 99.7%	≥ 99.9%
Median two-qubit gate (CZ) fidelity	≥ 98.0%	≥ 99.0%
Single-qubit gate duration	≤ 40 ns	≤ 20 ns
Two-qubit gate (CZ) duration	≤ 100 ns	≤ 60 ns
Median readout fidelity	≥ 95%	≥ 97%
Qubits in a GHZ state with a fidelity > 0.5	5	5
Quantum volume	≥ 8	≥ 16
Q-score	5	5
CLOPS_v	≥ 2400	≥ 3000

Click here for more

Physical infrastructure & operating conditions

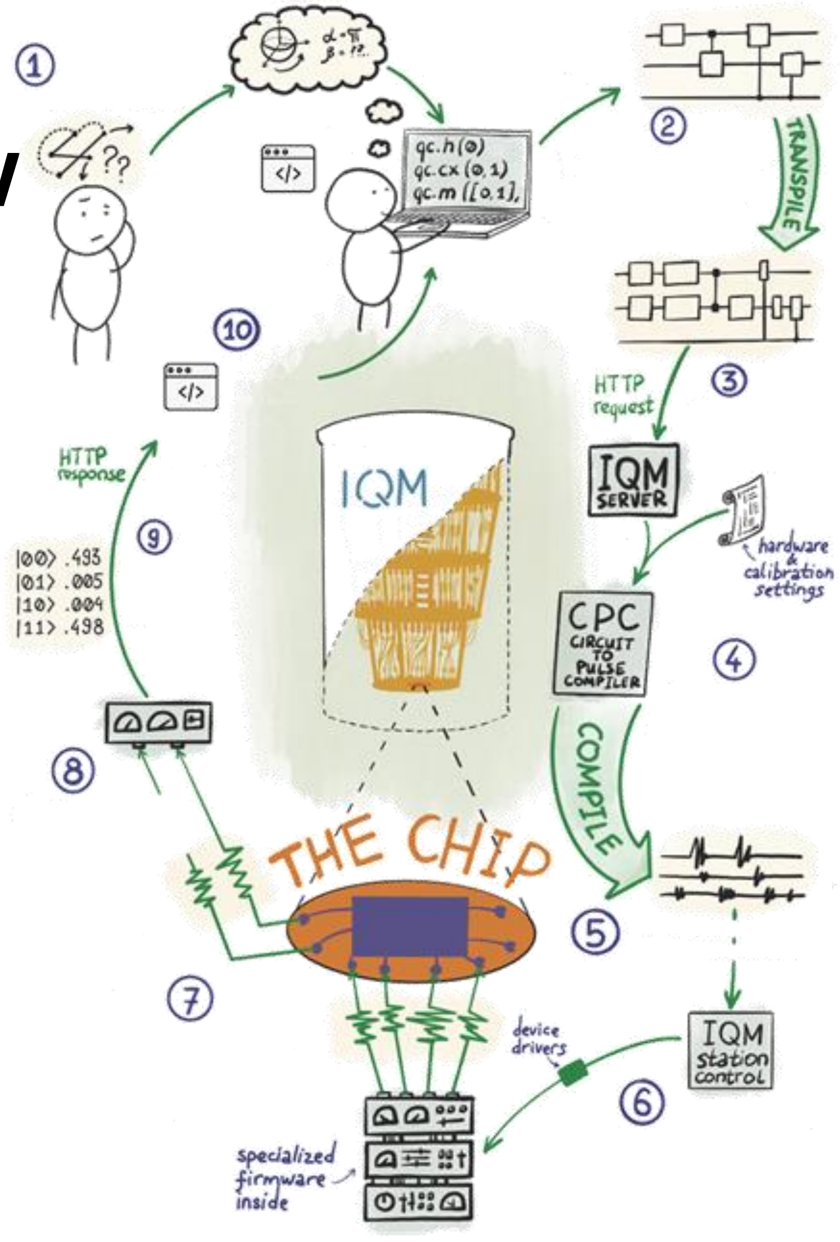
- To operate correctly, the QPU must be kept at around 20–30 millikelvin, that is –273 °C, practically absolute zero.

- This is achieved using a dilution refrigerator, a special cryostat that uses helium-3 and helium-4 to reach such temperatures.

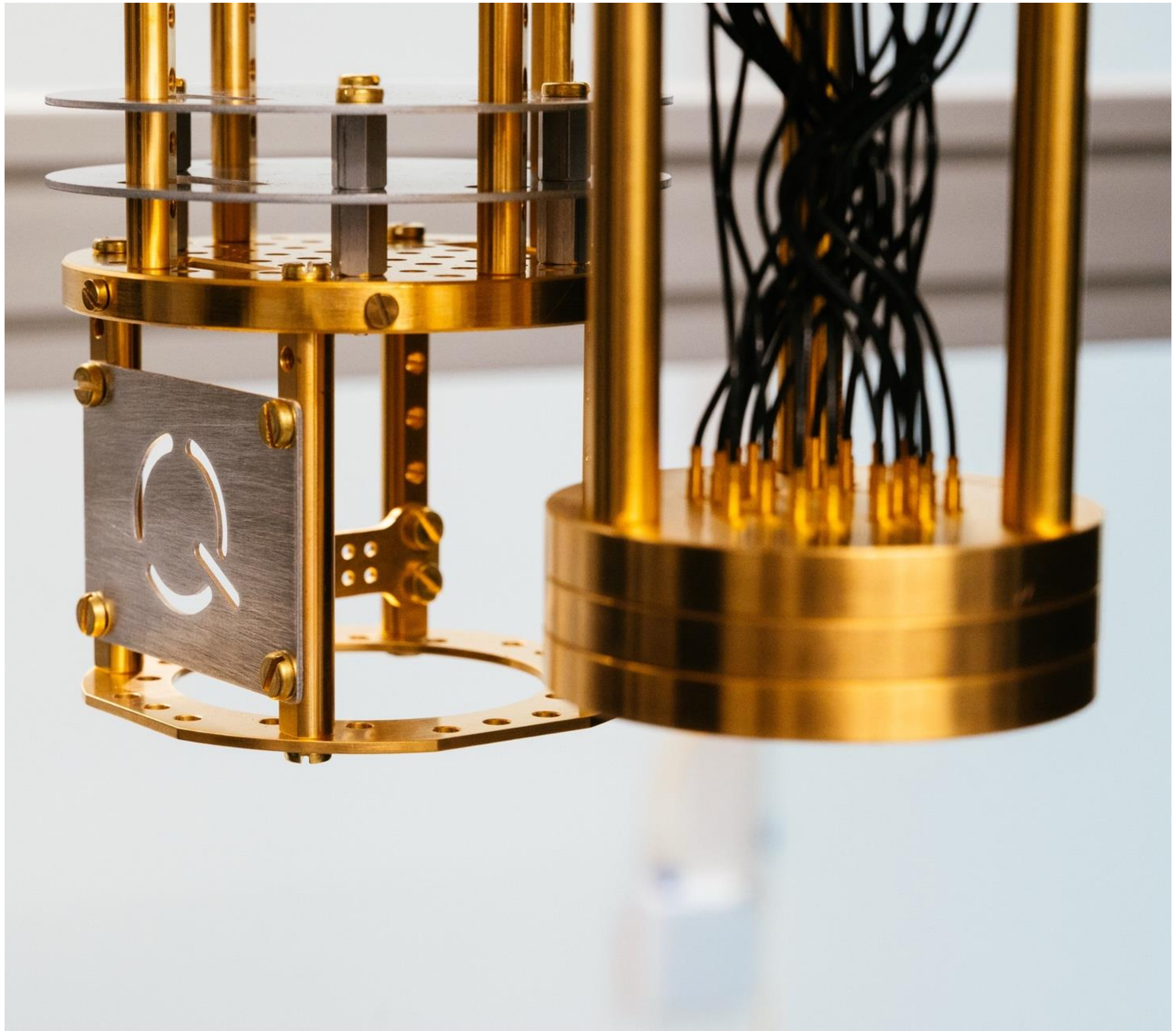
SOFTWARE

IQM Software Stack Overview

- Three main functional layers, each targeting a different interaction level:
  - **Cortex** – high-level execution of quantum circuits and **algorithms**
  - **EXA** – **experiment** orchestration, **calibration**, and pulse-level access
  - **IQM Station Control** – deterministic **low-level control** of instruments and **timing**
- Multiple entry points depending on the user's objective:
  - ☑ **algorithm** development and **benchmarking**,
  - ☑ device **calibration** and **characterization**,
  - ☑ hardware-level **control** and **signal** engineering.
- Provides a **closed end-to-end workflow**:
  quantum circuits → native gates → pulse schedules → physical signals → measurement data.
- Designed to support **research, education, and hardware–software co-design** on an on-premises quantum computer.



On-premises superconducting quantum computer for education and research, J. Rönkkö et al, EPJ Quantum Technology, April 2024



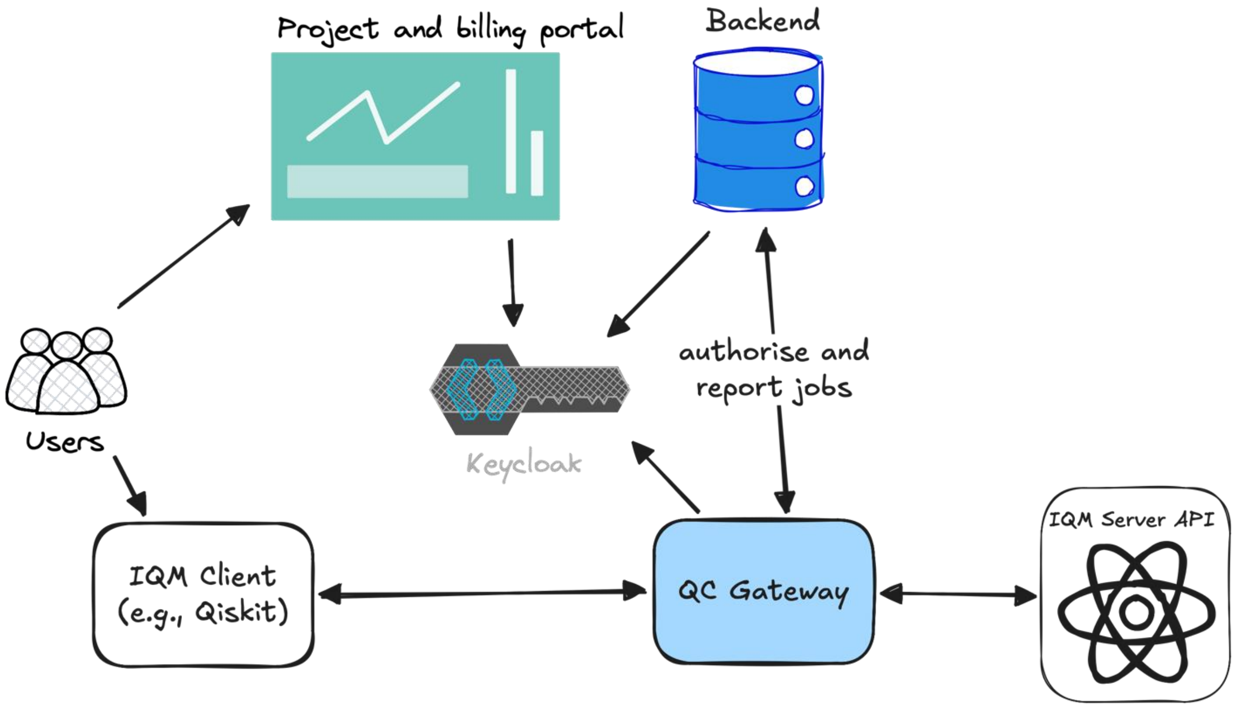
Access software

- IQM initially provided the machine with authentication only, **we needed a complex project billing/quota system** to manage several organizations, projects, users
- We needed HPC-like bookkeeping
  - **Projects, budget, billing, reservations**
  - Job authorisation based on these points
- And some more
  - SSO with federated institutional accounts

What we built

Overview

- **We built a flexible interface** to achieve that, **keeping IQM client compatibility** and integrating with existing identity infrastructure with
- An **API proxy** that
  - Authenticates the user
  - Asks authorization for the job submission to the back-end API
  - Collects the response with job_id for bookkeeping
  - **Fully transparent to IQM client**
- A **project dashboard**
  - A bit like cineca userdb + calendar reservations + jobs
- A **back-end API** that manages projects, jobs and provides authorization endpoints to the API proxy (based on budget, reservations, etc)
- A **login client** to support our SSO system, **compatible with iqm client**

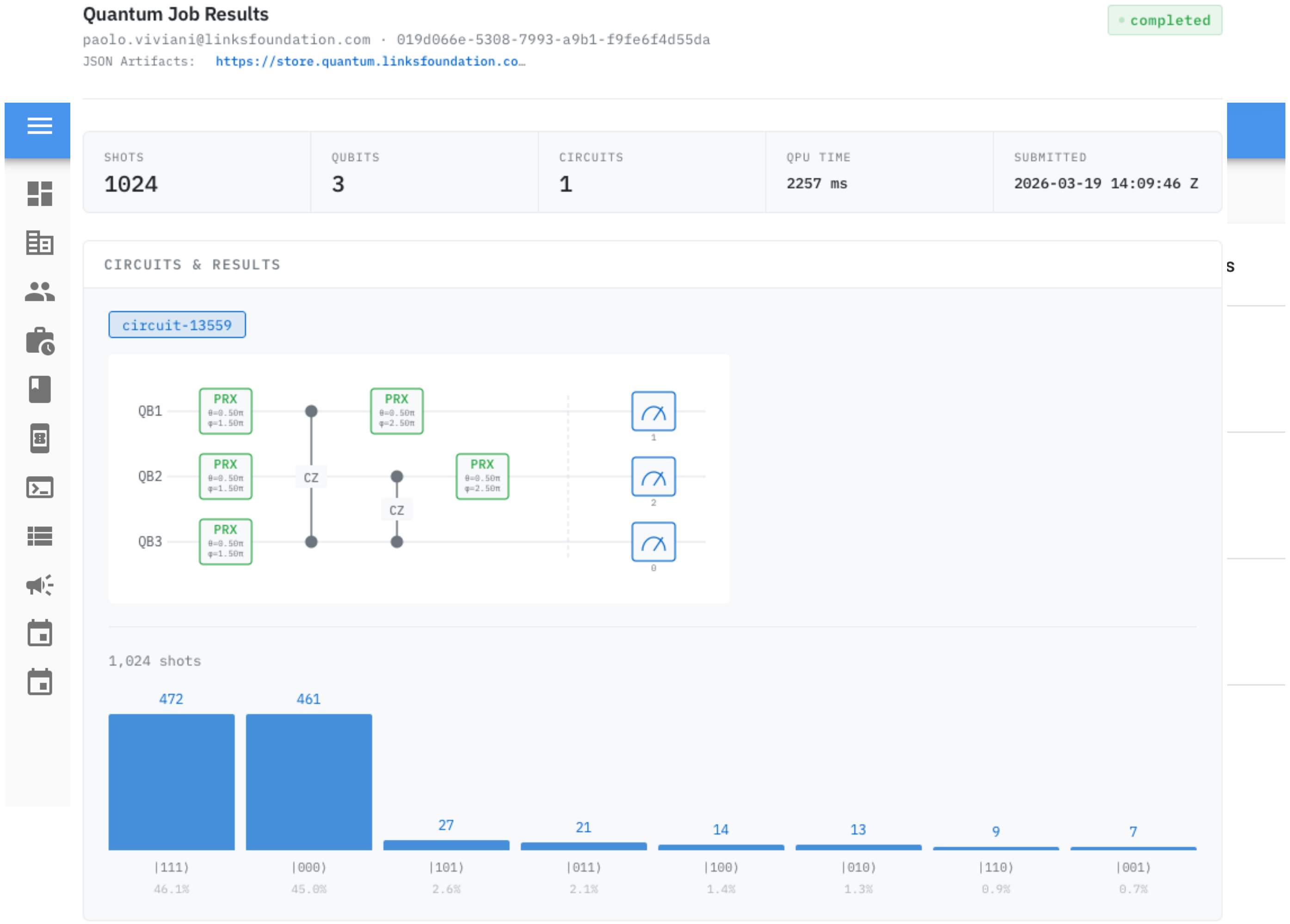


What we built

User interaction

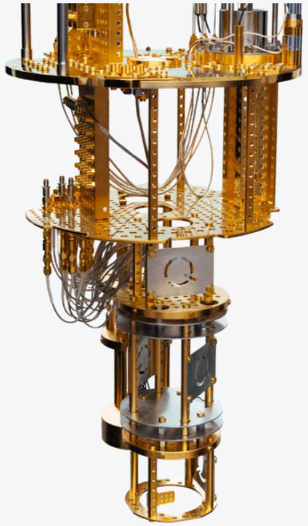
Users can

- **Submit jobs with IQM client** in a fully transparent way
- Check the machine status, occupancy and reservation calendar
- Check and manage their projects/budget
- **Access the results of their jobs** that are stored long-term
- **Access the calibration report** associated with each of their jobs



Quantum Computing and Simulation Lab - Lagrange Quantum Computer

CRYOSTAT

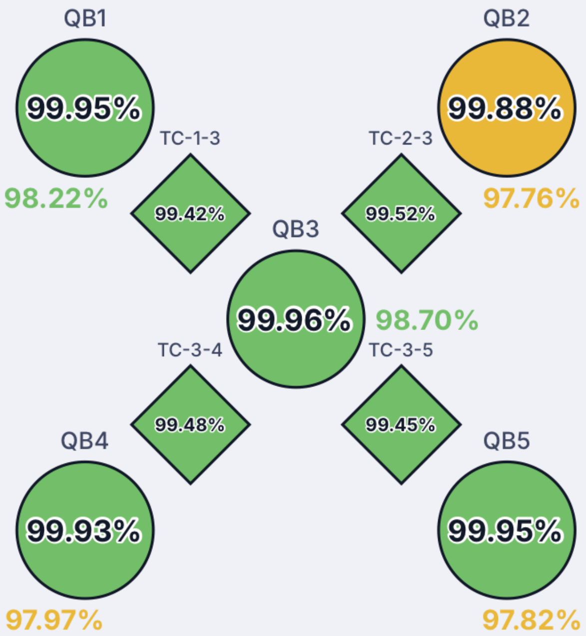


MXC temperature

0.02106 K

-273.13 °C

CALIBRATION



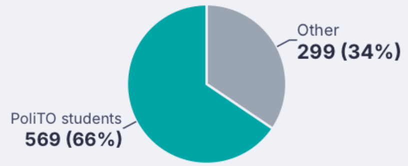
in qubit: PRX gate · outside: readout · in coupler: CZ gate

USAGE

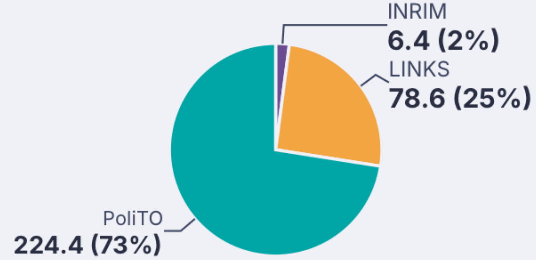
Numbers since September 2025

QPU usage (h)	309.4
Registered users	868
Jobs executed	775330

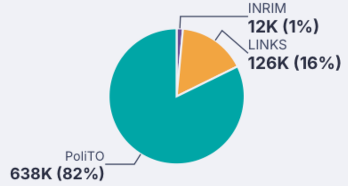
Users



Actual QPU time by organization (hours)

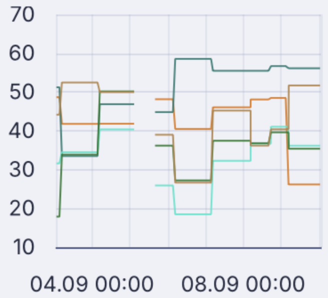


Quantum jobs executed by organization

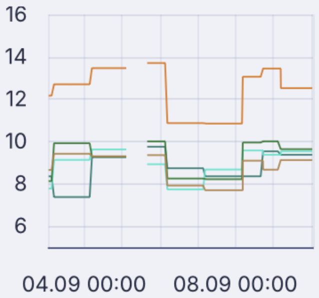


Compressor water

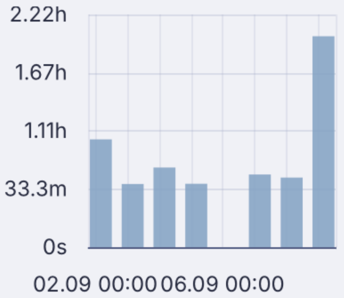
T1 time (µs)



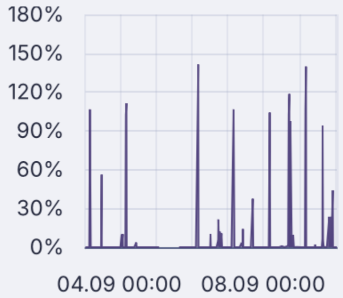
T2 time (µs)



Daily time of usage



QPU Occupancy rate



Take home messages

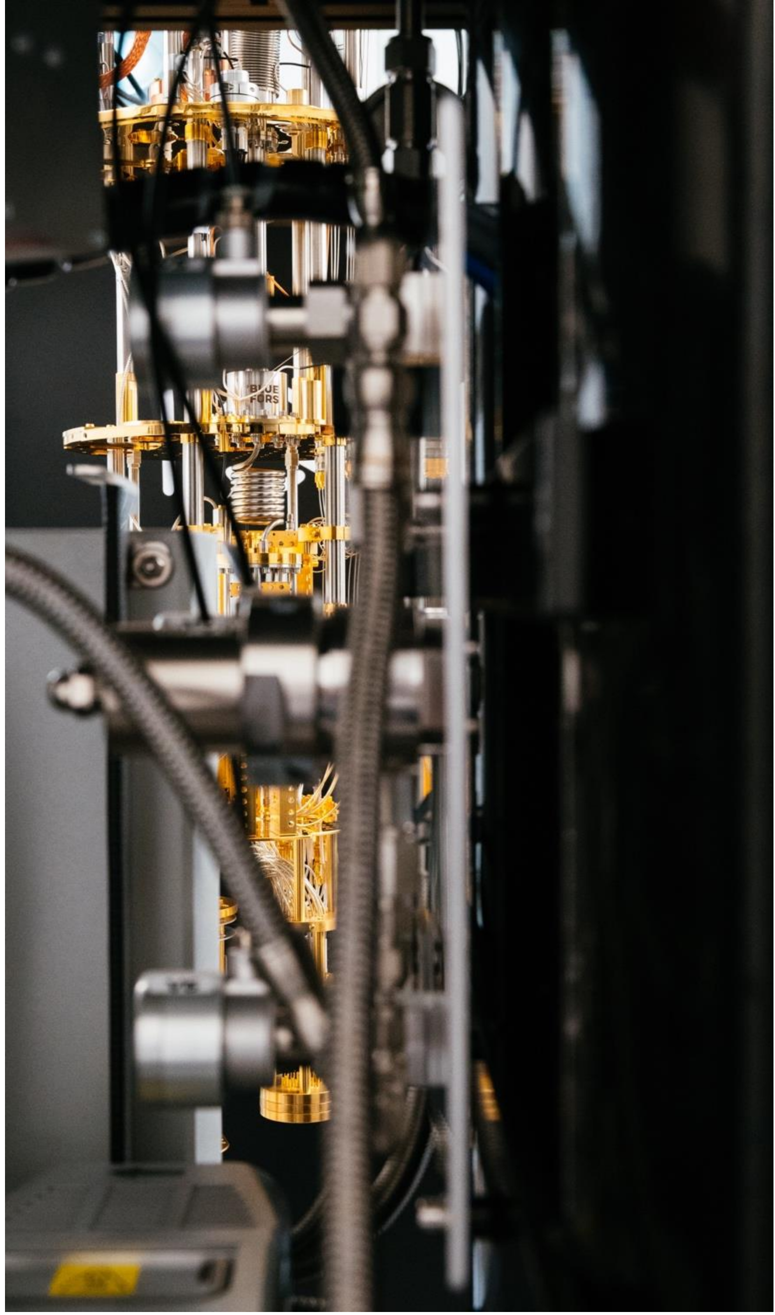
Transparent integration pays off

We built a transparent access middleware:

- No modification to the vendor client
- Handles authentication, quotas, fairness and complex billing policies for several hundreds users and hundreds of thousands of jobs since September 2025
- Beyond our Spark: the software is modular, we are discussing integration with SLURM
- **Now open source!** (github.com/LINKS-Foundation-CPE/QC-Gateway, reference paper arxiv.org/abs/2604.21695)

Code

Paper



Politecnico di Torino

FONDAZIONE LINKS PASSION FOR INNOVATION

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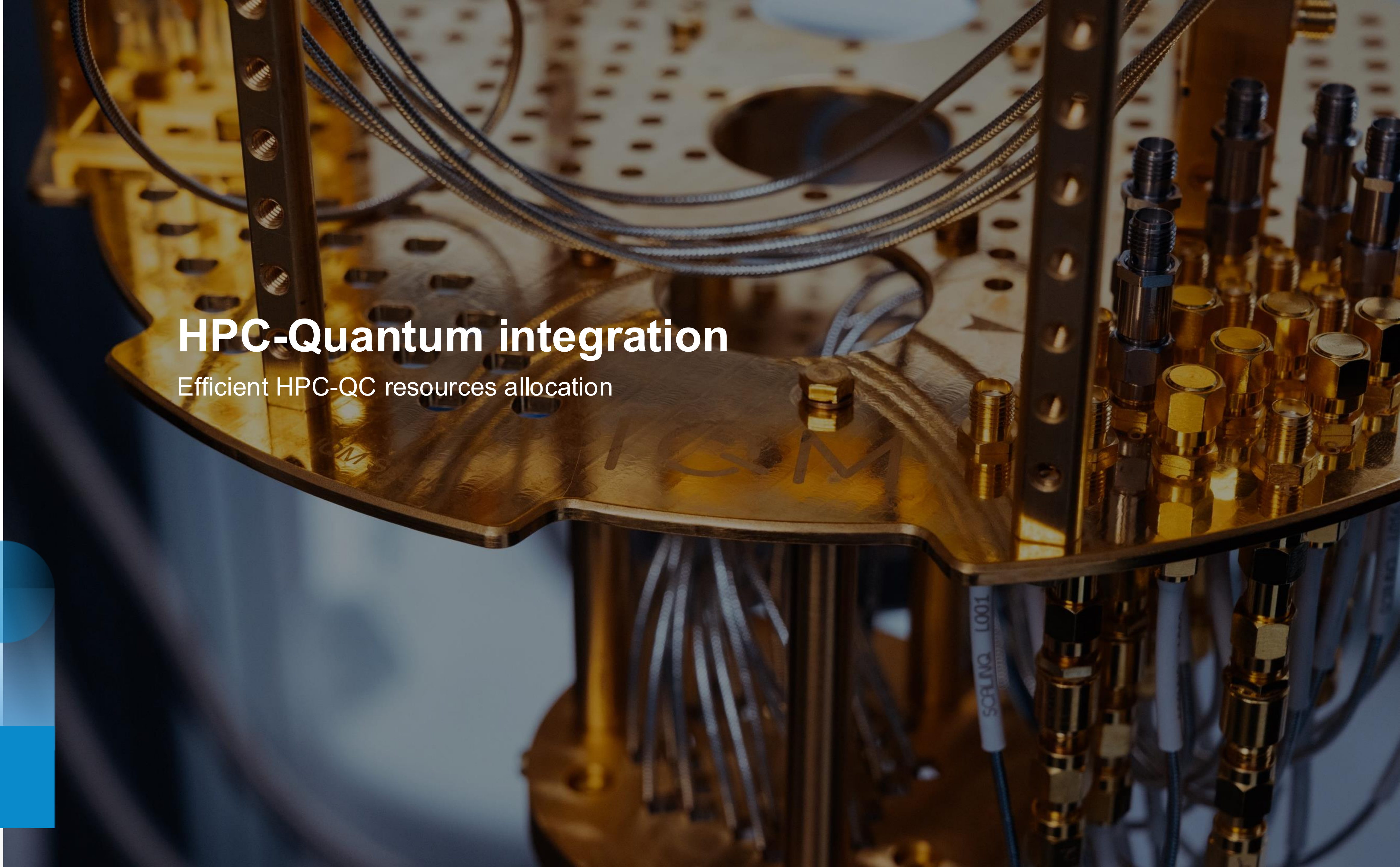


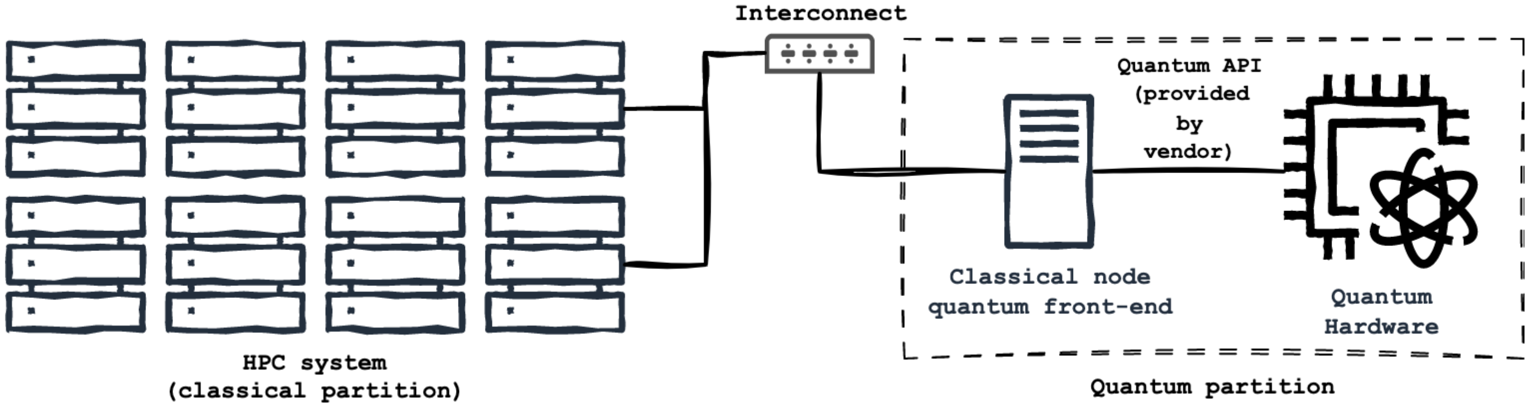
HPC-Quantum integration

Efficient HPC-QC resources allocation

SmartHPC.QC

National HPC-Quantum integration project

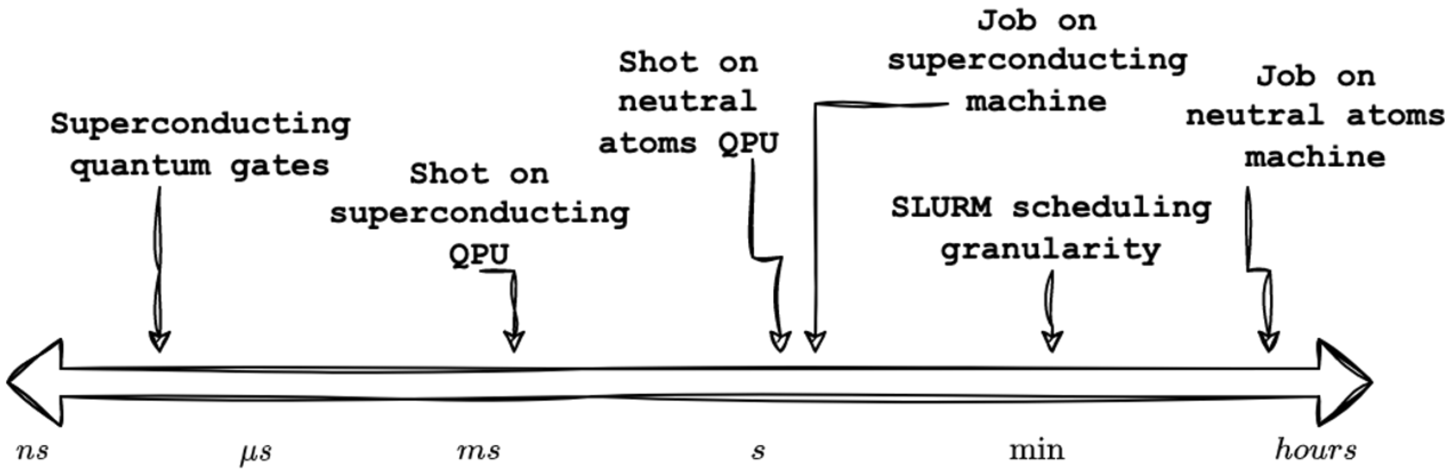
- The goal of the project is to define strategies for **HPC-Quantum resource scheduling** that work across different technologies (e.g., superconducting, neutral atoms, etc.), and **fit with HPC centre requirements**
- We identified **three main strategies** for different scenarios: **virtual QPUs**, **MPI malleability**, and **workflows**. We are focusing here on vQPUs.
- We are assuming a reference architecture like the following.



SmartHPC.QC E4 COMPUTER ENGINEERING FONDAZIONE links PASSION FOR INNOVATION CINECA UNIVERSITÀ DI TORINO BSC

A granularity problem

time scales for reference



- superconducting quantum gates: ns
- shot on superconducting HW: 1-10 ms
- shot on neutral atoms machine: 0.3-1 s
- **job on superconducting HW (~1000 shots): 1-20 s**
- typical SLURM scheduling granularity: ~min
- **job on neutral atoms machine (custom register preparation + 1000 shots): 15-30min**

HPC-QC integration

Co-scheduling considered harmful

EXAMPLE JOB :

- co-scheduling 10 nodes in a classical partition and 1 QPU (defined as a SLURM gres) in a quantum partition, both for 1 hour.
- 1) superconducting QPU: short quantum tasks (~ 10s) → possible heavy under-utilisation of the QPU.
- 2) neutral atoms QPU: long quantum task (~30min)→ possible under-utilisation of the classical nodes.

↓

simple co-scheduling with exclusive QPU access is inadequate for achieving optimal resource utilization in heterogeneous HPC-QC environments.

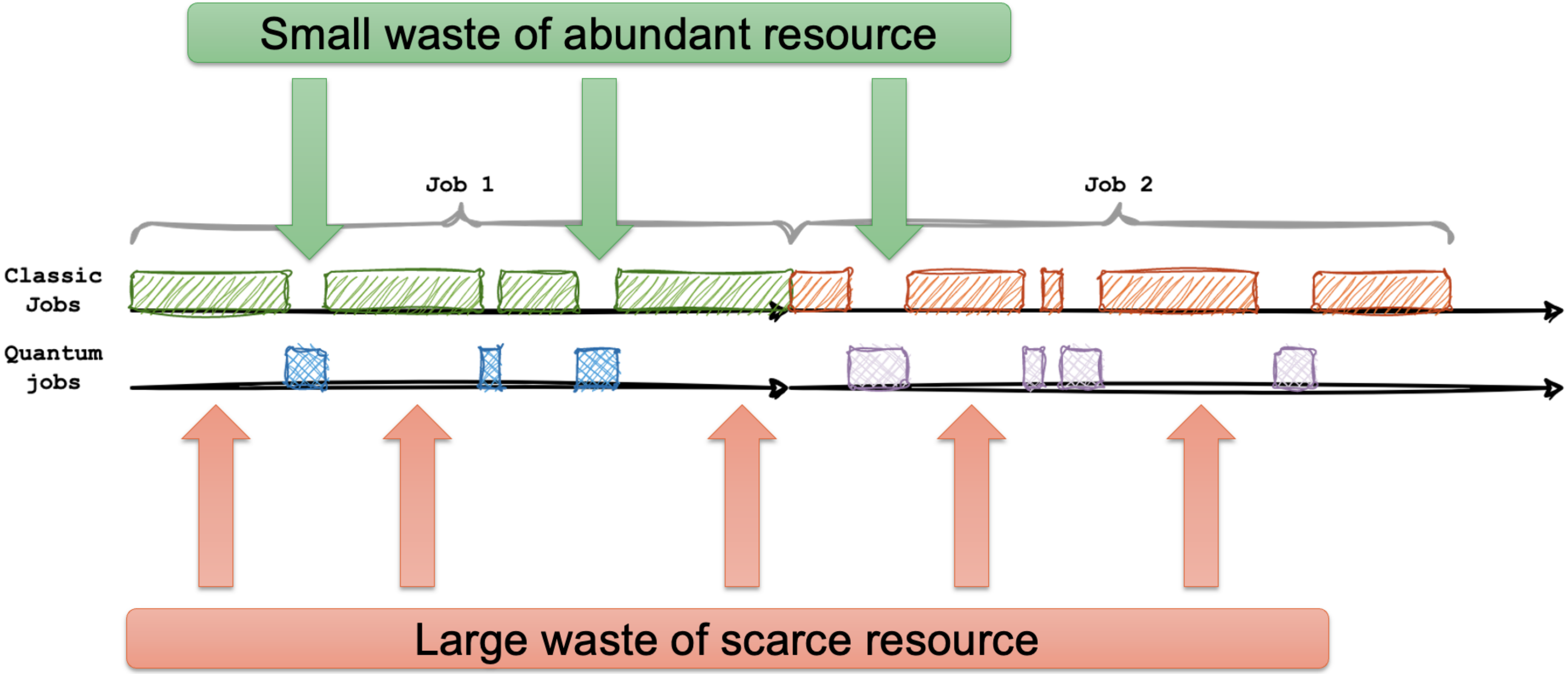
```
#!/bin/bash
#SBATCH --partition classical
#SBATCH --nodes 10
#SBATCH --time=01:00:00
#SBATCH hetjob
#SBATCH --partition quantum
#SBATCH --gres=qpu:1
#SBATCH --time=01:00:00

srun ./hybrid_job
```

Long classical, short quantum jobs

Focus on superconducting

Superconducting-like scenario

co-scheduling considered harmful

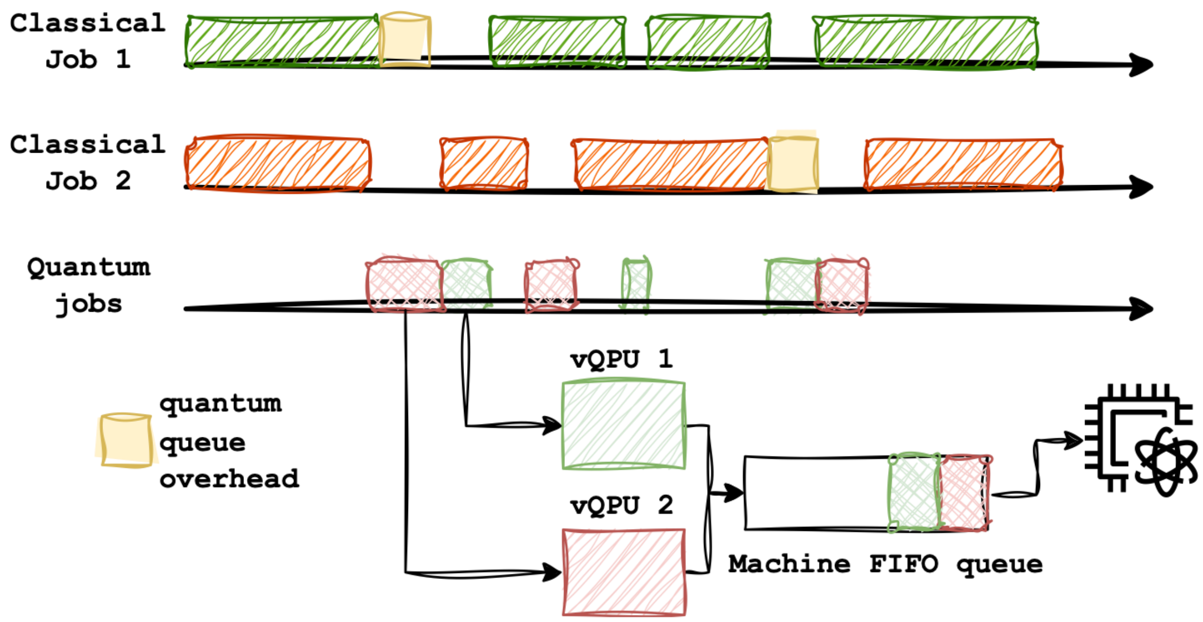


vQPU approach

sharing the QPU by time-multiplexing

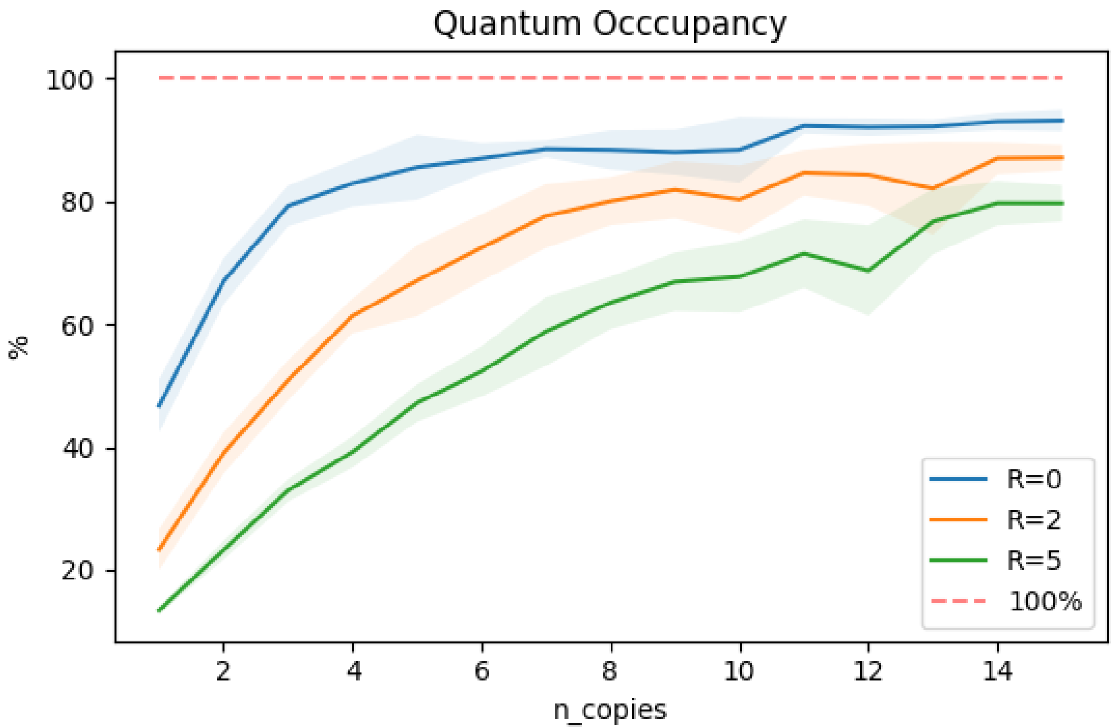
Proposed approach:

- We expose more QPUs than actually available through SLURM GRES, allowing for multiple jobs to access the QPU concurrently.
- The QPU still processes quantum tasks sequentially, but the latency is absorbed by:
  - 1) sparse quantum task execution
  - 2) small penalty on overall execution time



Quantum Occupancy

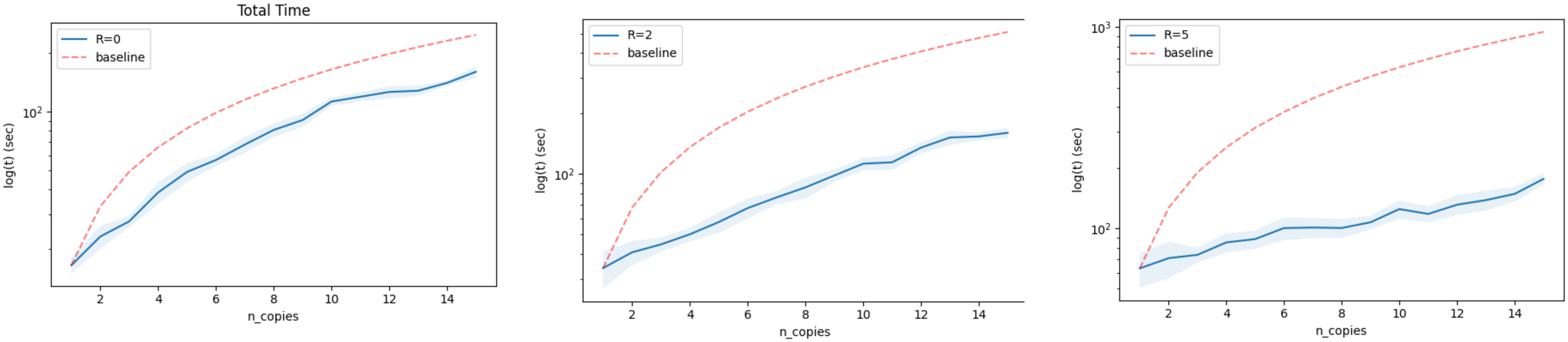
- *Quantum occupancy* is the percentage of occupation of the QPU during the execution of all workflows
- At the increase of the number of workflows, this percentage increases as well, until a saturation level is reached, due to the impossibility for the QPU to fill further "blank spaces" coming from the classical computations of the various GC job instances



Total time

Increasing the workload imbalance

- *Total time* is the time interleaving between the start of the first workflow and the end of the last one
- We observed a distinct performance gap between the vQPU and the baseline approaches
- By increasing the classical-quantum workload imbalance of each job, the performance gap between the two resources scheduling strategies enlarges as well
- This proves vQPUs to be a well-suited approach to handle hybrid jobs characterized by such a disparity



The advantage of vQPUs

- The experimental results demonstrate that time-based multiplexing optimizes QPU usage, while significantly reducing overall execution time at cluster level

- In addition, this performance gain becomes increasingly pronounced as the workload imbalance grows, making this approach particularly effective under the considered superconducting-hardware scenario

- The observed trends also suggest that increasing the number of vQPUs instantiated in the HPC system does not significantly affect the average job queue time past a certain point, indicating that this strategy remains effective under varying HPC workloads

- This work includes one of the **very first examples of HPC-Quantum hybrid jobs fully executed in Italy**

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Three ways to share a QPU: Scheduling strategies for hybrid Quantum-HPC applications[☆]

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ABSTRACT

As quantum computing (QC) technologies mature, their integration into established high-performance computing (HPC) infrastructures is becoming a central objective for next-generation computing systems. However, unlocking the potential of hybrid platforms for computationally demanding workloads remains challenging. The mismatch between quantum and classical programming models, the limited maturity of quantum software stacks, and the scarcity of quantum processing units (QPUs) above all, necessitate scheduling strategies that go beyond standard HPC mechanisms to manage such heterogeneous and constrained resources.

To address this issue, we investigate three distinct methodologies for HPC-QC resource scheduling: time-based multiplexing, dynamic resource management, and workflow decomposition. Experimental validation on production HPC clusters and real quantum hardware demonstrates the effectiveness of these approaches under different workload scenarios. Malleability and workflow strategies significantly optimise classical resource utilisation, reducing consumption by up to 45.7% and 64% respectively, proving to be best fitted for hybrid jobs where quantum and classical workloads are evenly balanced. Conversely, time-multiplexing enhances QPU utilisation and reduces execution time at the cluster level, making it the optimal strategy for the opposite context, which is characterised by high classical-quantum workload imbalances. These findings underscore the practical viability of tailored scheduling strategies for hybrid HPC-QC environments and highlight their complementarity in building efficient, scalable software stacks for next-generation quantum-accelerated facilities.

1. Introduction

Quantum computing (QC) is evolving rapidly, strongly establishing itself as a cornerstone in the future of advanced technologies. In this context, integration with traditional high-performance computing (HPC) systems is regarded as a crucial step towards fully exploiting its potential [1,2], leveraging the unique capabilities of QC to accelerate computationally intensive and exponentially scaling workloads while leaving the rest to the classical system. However, such integration raises new challenges on both sides, potentially hindering the development of both if left unchecked.

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[1] Equal contribution.

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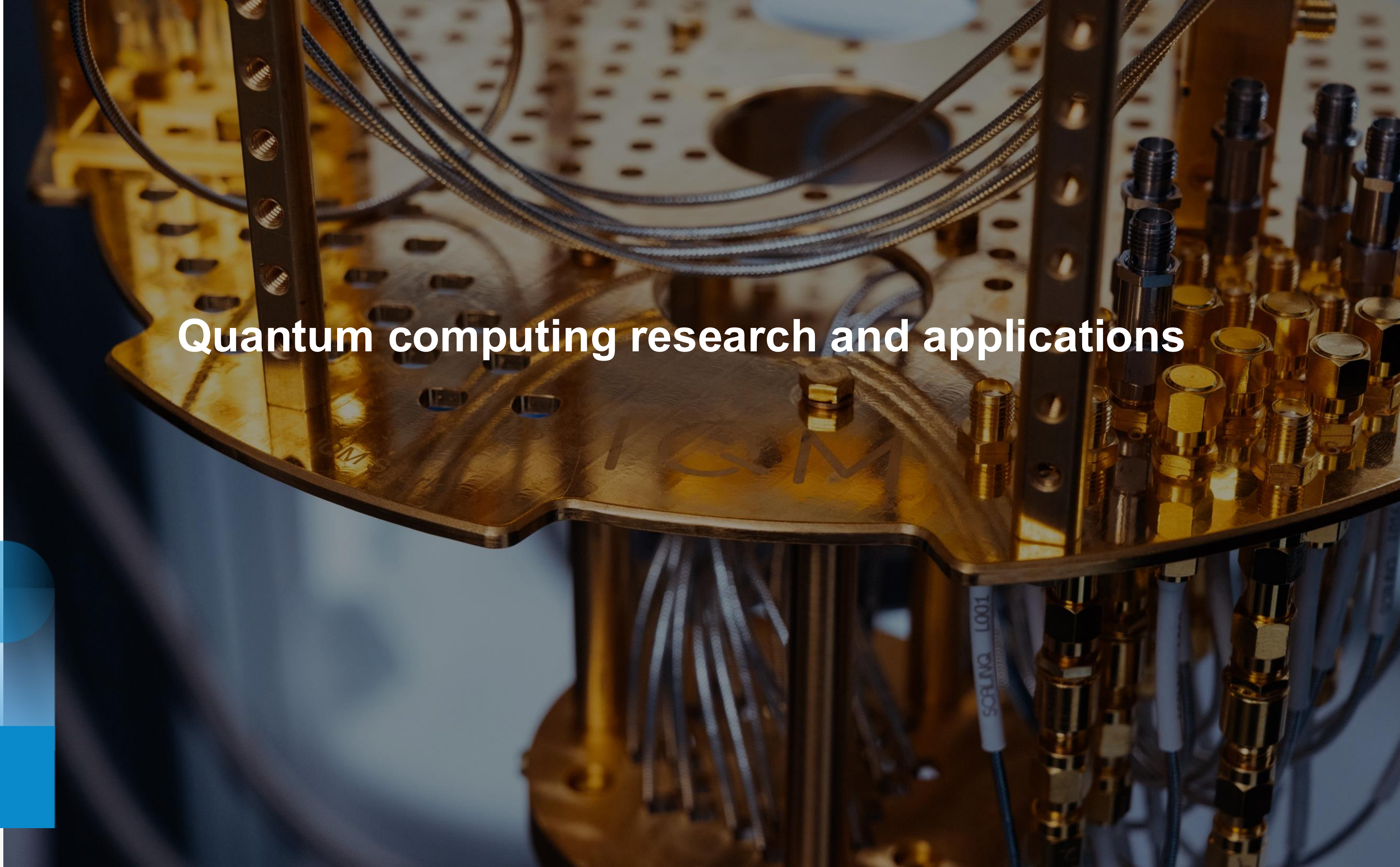
Looking for more detail?

Three ways to share a QPU

Quantum computing research and applications

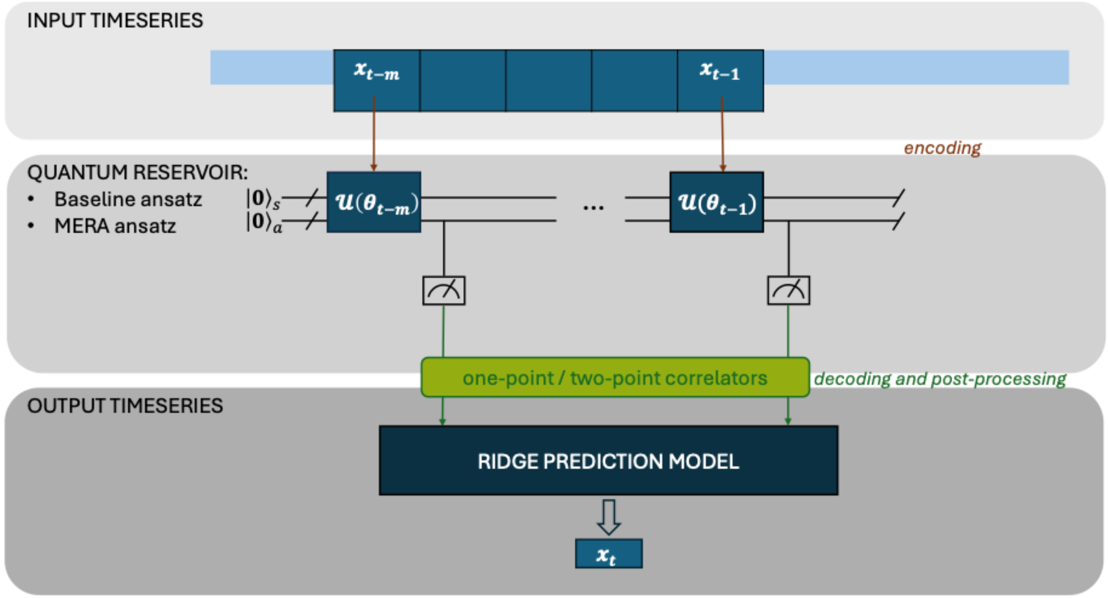


Time-Series Prediction using QRC on IQM Machines

Description

Objective

- Apply Quantum Machine Learning (QML) to financial time-series forecasting
- Evaluate the Quantum Reservoir Computing (QRC) paradigm on real IQM Spark hardware
- Compare quantum approaches (MERA, RM) against the classical baseline Prophet (Meta)



Context

- NISQ era (Noisy Intermediate-Scale Quantum): real devices with noise and hardware constraints
- Dataset: ATM transactions (84 time steps per ATM, split 64 train / 20 test)

Time-Series Prediction using QRC on IQM Machines

Methodology

Dataset

- ATM transactions (Kaggle): total balance per ATM
- 84 time steps per ATM (64 train, 20 test)
- Iterative multi-step forecasting (sliding window)

Classical Baseline: Prophet

- Additive model (Facebook): trend + seasonality + error
- Robust to missing data and outliers

Execution Backends

- AerSimulator (ideal, noiseless)
- IQM FakeAdonis (simulation with realistic noise)
- **IQM Spark (real quantum hardware, 5 qubits)**

Quantum Models (QRC)

MERA (Multiscale Entanglement Renormalization Ansatz)

– Hierarchical multi-scale tensor network architecture

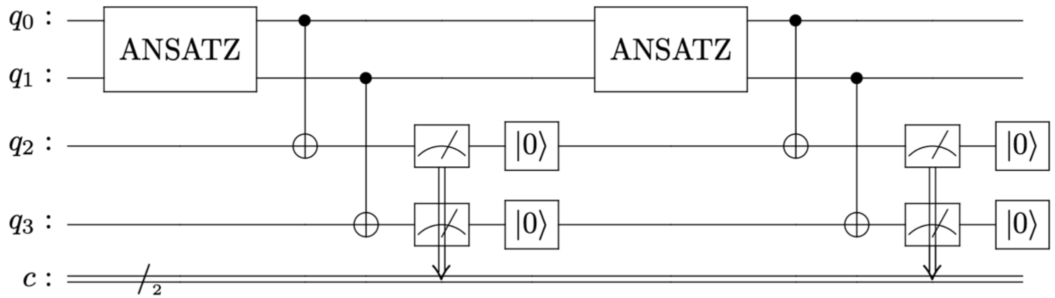
– Best performance on short-term forecasting

– Sensitive to excessive memory (noise)

RM (Repeated Measurement)

– Repeated measurements extend the temporal memory of the reservoir

– More effective on longer horizons and on real hardware



Time-Series Prediction using QRC on IQM Machines

Results

QRC vs. Classical Approach

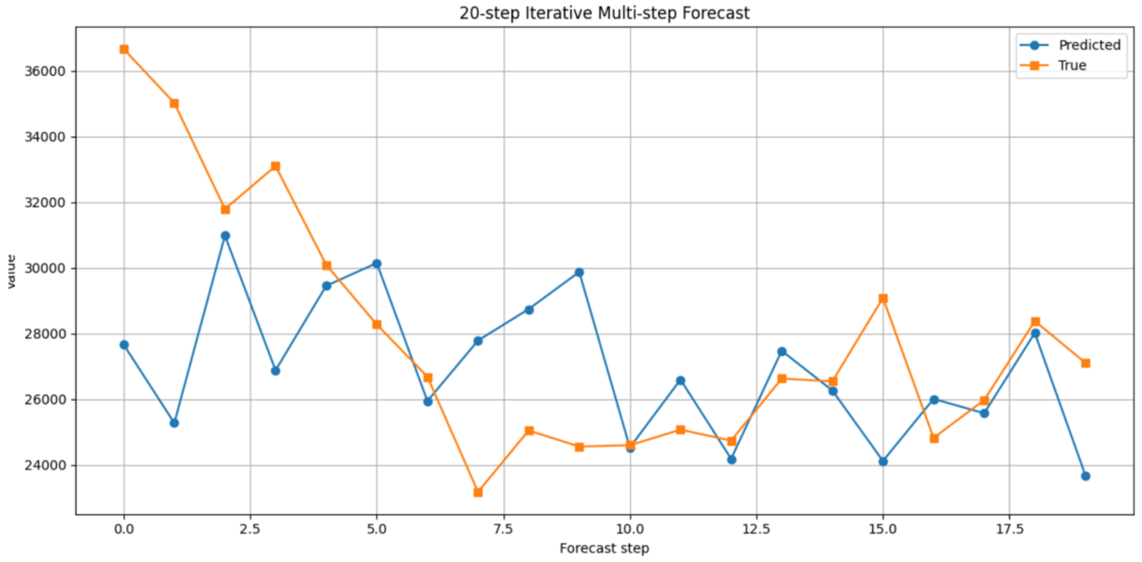
- QRC models (MERA and RM) **comparable to Prophet** across all key metrics (MAE, MAPE, MSE, RMSE)
- High-dimensional Hilbert space provides richer encoding of temporal dependencies

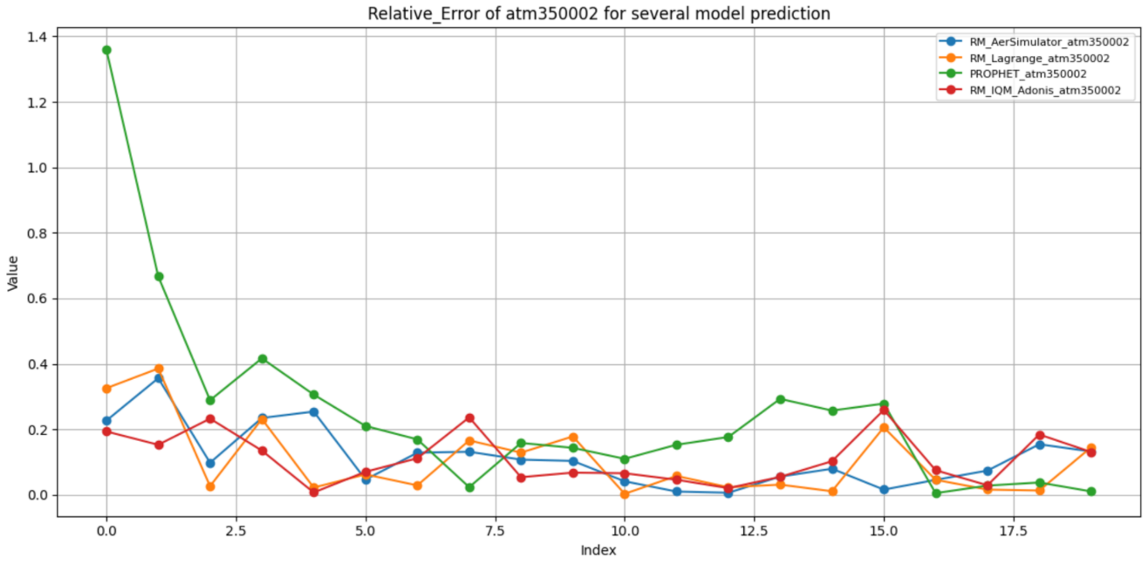
MERA vs. RM — Complementary Behaviour

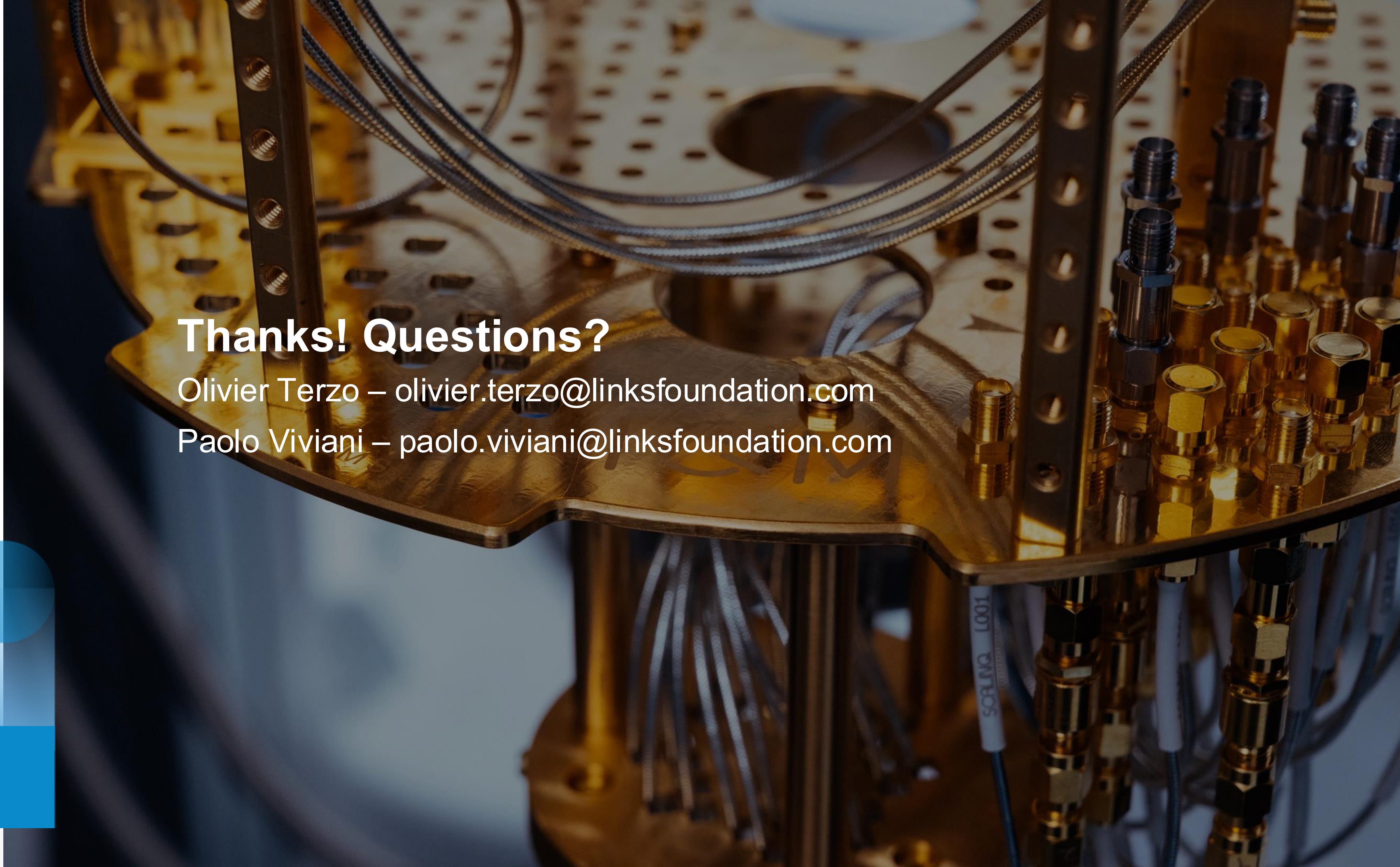
- MERA: more accurate short-term (first 10 steps); RM: superior long-term and on IQM Spark

Limitations & Future Work

- Small dataset and deep circuits are the main constraints in the NISQ era
- Next steps: larger datasets, hybrid MERA+RM architecture, tests on additional quantum hardware







Thanks! Questions?

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